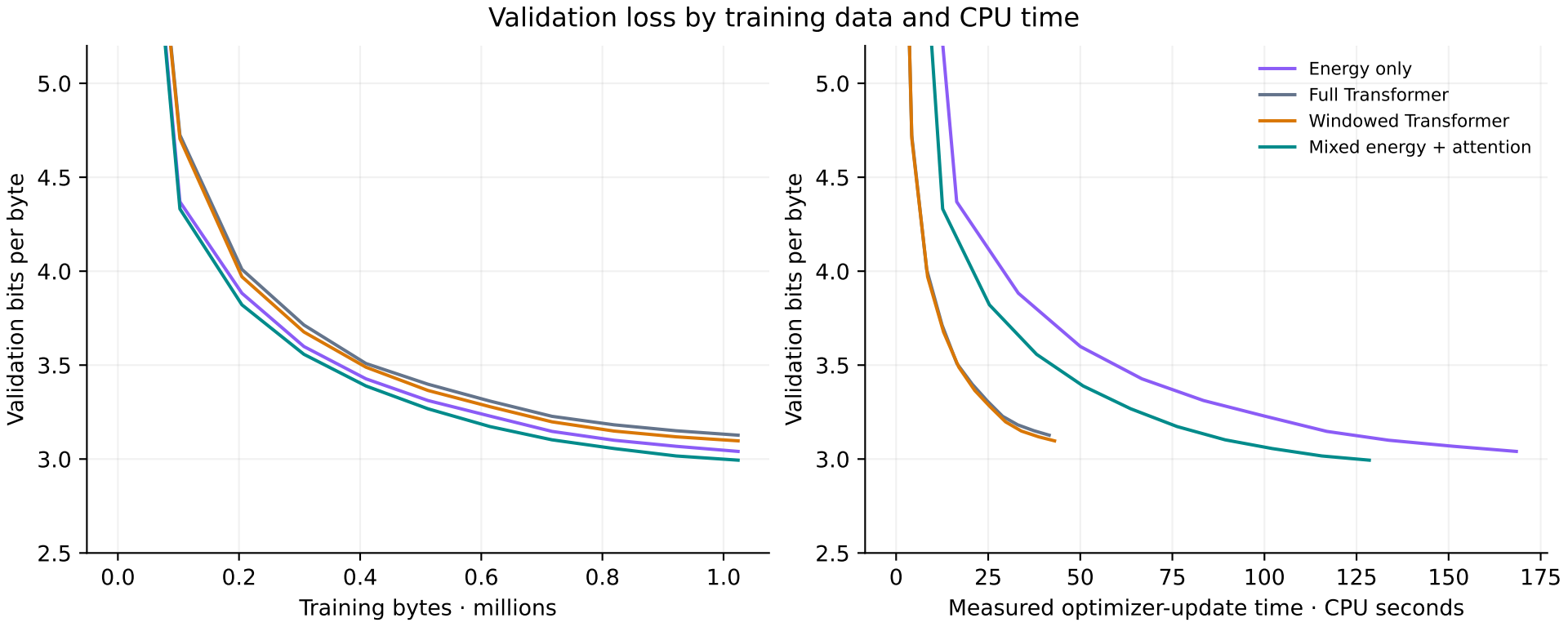
Validation loss by training data and CPU time
Validation bits per byte
5.0
4.5
4.0
3.5
3.0
2.5
0.0
0.2
0.4
0.6
0.8
1.0
Training bytes · millions
Validation bits per byte
0
25
50
75
100
125
150
175
Measured optimizer-update time · CPU seconds
Energy only
Full Transformer
Windowed Transformer
Mixed energy + attention